Response Letter with Annotations

Witte, M., Heitmann, M., Hartmann, J., & Tetzlaff, K. (2026). Language of images: Classifying marketing images with transformers and vision language models. *International Journal of Research in Marketing,* S0167811626000017.

https://doi.org/10.1016/j.ijresmar.2026.01.001

Annotations

1. **Purposes**
   - **a.** Motivation: So far, no large-scale comparison between VLMs and TVMs exists, which makes the benefits and limitations compared to CNNs unclear. A better understanding is important because marketing can be tempted to confuse impressive lighthouse performances with overall reliability.
   - **b.** Specific Rationale (several bullet points)
     - Prior studies focused on:
       1. Convolutional neural networks (CNN) have advanced marketing's ability to classify images and to understand and manage visual impacts.
       2. However, CNNs have limitations in terms of accessibility and accuracy.
       3. CNNs work well for classifying what is visible
       4. How consumers perceive an image often depends heavily on context and image configuration (Heitmann et al., 2025); which CNNs might not capture in full detail.
     - Need to:
       1. Evaluate alternative methods and paradigms from an applied perspective, presenting empirical evidence on accuracy and practical guidance for model choice in marketing
       2. Recently emerged transformer-based vision models (TVMs) include conceptual solutions. Rather than simply grouping pixels in an unordered fashion, TVMs follow techniques from text analysis, where relationships between textual elements matter and create meaning. TVMs weight the joint occurrence of image elements accordingly.
       3. Vision language models (VLMs) like OpenAI's GPT-5 and Microsoft's Phi-4 (Abdin et al., 2024; OpenAI, 2025a), also known as multi-modal large language models, might better capture such visual-emotional dynamics.
          a. By representing visual and linguistic information in joint multi-modal layers of abstractions VLMs relate visuals and

rich textual descriptions more intricately than CNNs and TVMs

b. Unlike conventional CNNs and TVMs, that both require adapting neural network weights to link image signals with novel labels, applying pretrained VLMs require neither AI training knowledge from users nor computational training resources. They also do not require manually annotated training images, which can be complex and costly to collect, in particular when many label categories exist or variance in subjective perceptions requires multiple annotations

c. Research Questions and/or Hypotheses

- How effective are VLMs compared to CNNs or TVMs in actual marketing-related applications?
- Which models should marketing rely on?
- In this research, we will therefore investigate whether similar performance differences between paradigms exist in image classification, and to what extent lighthouse illustrations of recent image classification methods generalize to various applied marketing tasks and datasets.
- **Identify IV, DV, mediator, moderator**
    1. **IV1:** CNN (InceptionV3, ResNet152, Xception, and CNN trained from scratch.
    2. **IV2:** TVM (ViT, ConvNeXt)
    3. **IV3:** VLM (GPT-5 and Phi-4).
    4. **DV:** Average performance across datasets
    5. **DV2:** Performance on dataset level
    6. **DV3:** Performance across image-related task types
    7. **DV4:** Accuracy of image classification ensemble methods
    8. **Moderators:** Large-data and limited-data scenarios
- Identify specific relationships among variables
    1. CNN → accuracy of image classification paradigms
    2. TVM → accuracy of image classification paradigms
    3. VLM → accuracy of image classification paradigms
    4. Conditional effect (large vs limited data scenarios) → accuracy of image classification paradigms

2. Method

a. Research Design

- Computational study

b. Sample

- Evaluate 9 methods (5 CNNs, 2 TVMs, 2 VLMs) under large and limited training data conditions, including 9 hyperparameter optimization runs across 18 data sets.

- This results in 2275 total experiments (7 fine-tunable methods x 18 datasets × 2 data availability conditions × 9 hyperparameter optimization runs + 2 methods without task-specific fine-tuning x 18 datasets × 2 data availability conditions).
- Measure accuracy as the primary performance metric because it is the most widely reported metric in marketing publications.

c. Measures

- **Training:**
    1. Randomly sample 1,000 images per dataset (large training data) and also limit training data to only 200 images per dataset (limited training data), reflecting typical training data sizes in marketing.
- **Accuracy of image classification paradigms**
    1. Average performance across datasets
    2. Performance on dataset level
    3. Performance across image-related task types
    4. Accuracy of image classification ensemble methods

d. Analysis

- Constructed a unified hold-out set that is shared across all training conditions
- Hyperparameter optimization for each method, dataset, and training data size combination
- Computed global average accuracy across all datasets to establish an expected benchmark accuracy for comparison.
- Implement an ensemble of the best performing VLM and TVM, which adds 315 additional experiments to the analysis.

3. Results and Conclusions

a. Answers to RQs and Hypotheses

- When training data is limited, average accuracy of GPT is about 74%. This is better than TVMs like ConvNeXt (66%) and ViT (61%), which make 8% and 13% more mistakes when assigning labels.
- Differences diminish when more training data is available.
- The open-source VLM Phi does not reach GPT performance on average, with a sizeable difference of about 12% accuracy.
- Among CNNs, Xception is consistently best performing at 61% (limited training data) and 69% (large training data). However, none of the CNNs reach the performance of TVMs and VLMs.
- For both TVMs and CNNs, adding training data pays off with average gains in accuracy between 8% and 12%.
- In Figure 4, the various intersections between the performance lines demonstrate that relative performance depends on the image classification tasks and datasets.
- VLMs can perform on par with top-performing TVMs for what and how tasks, despite requiring neither fine-tuning nor additional training data.
- In contrast, VLMs perform poorly on the who-tasks that marketing research has examined thus far. Results from datasets related to ethnicity

or gender detection indicate that this underperformance likely stems from ethically motivated restrictions designed to prevent potential misuse.

- TVMs and VLMs not only perform well on distinct types of visual data but also capture complementary representational features.

b. Include Key Figures/Tables

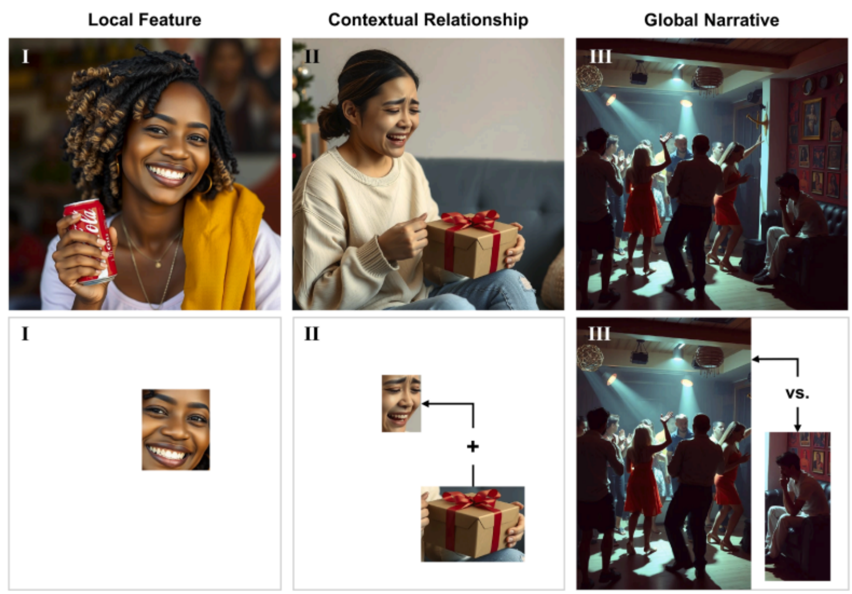


Fig. 1. Image classification scenarios with increasing difficulty. *Note:* Upper panel shows original images (FLUX-generated). The lower panel indicates key image regions and relationships needed for correct classification.

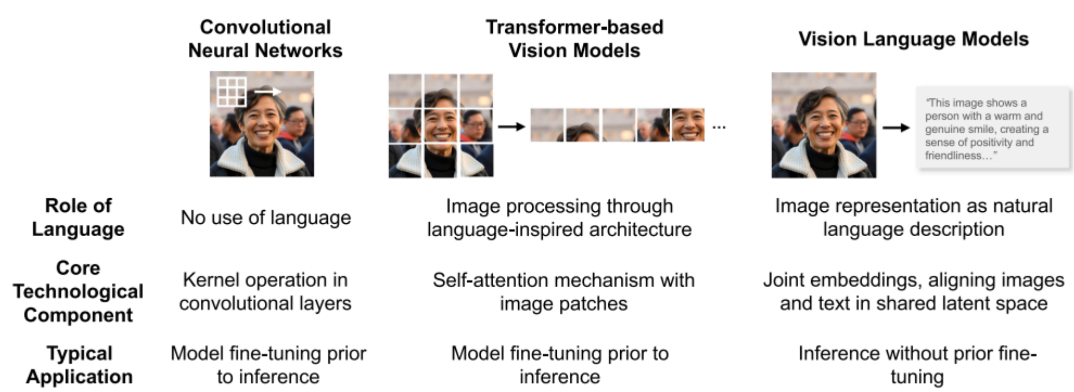


Fig. 2. Overview of image classification paradigms.

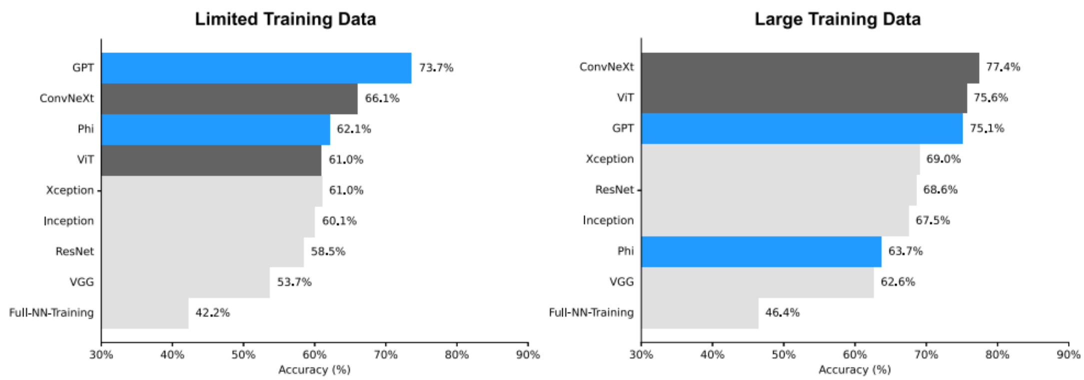
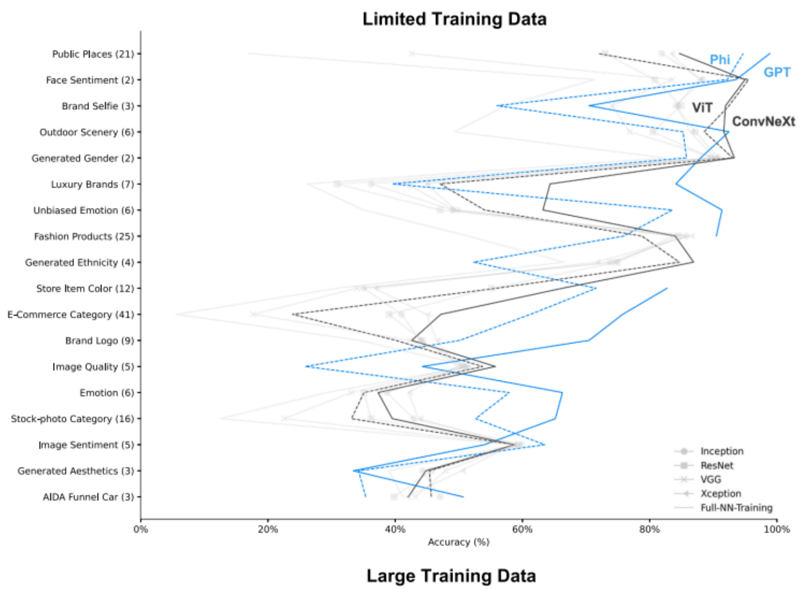


Fig. 3. Average accuracy across all datasets across methods. *Note:* Performances of CNNs are displayed in light gray, TVMs in dark gray, and VLMs in blue.

M. Witte, M. Heitmann, J. Hartmann et al.
International Journal of Research in Marketing xxx (xxxx) xxx



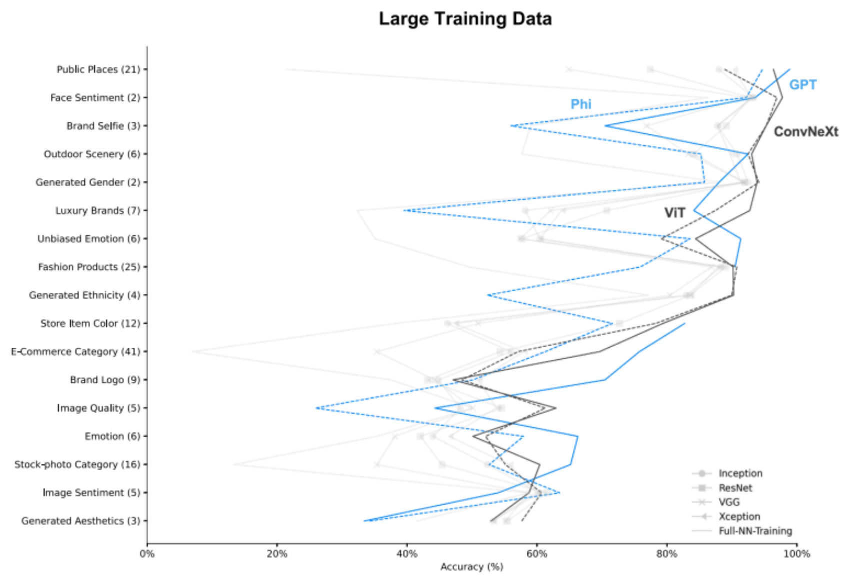
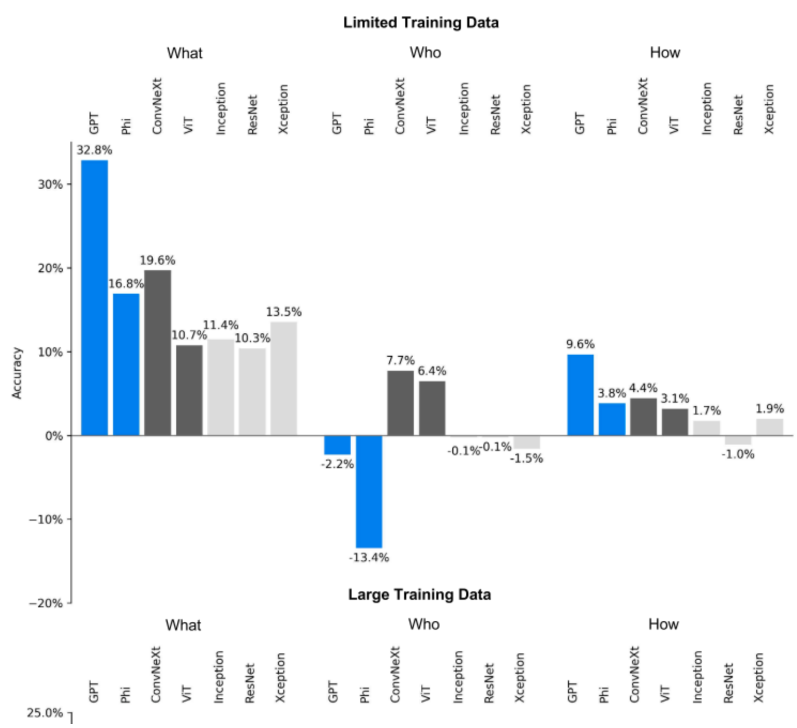


Fig. 4. Method performance per dataset. *Note:* Number of classes is provided in parentheses to the right of each dataset name. Blue solid line represents GPT, blue dotted line Phi. The black solid line stands for ConvNeXt and the black dotted line for ViT.



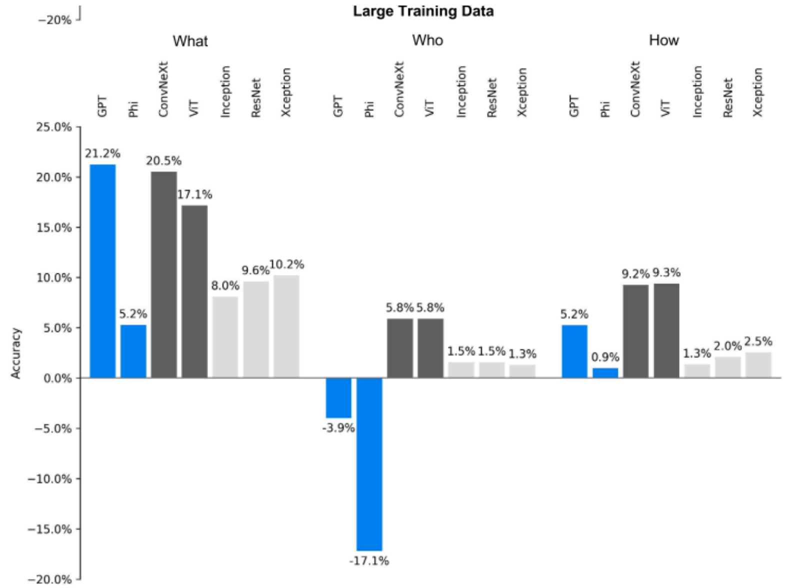


Fig. 5. Accuracy depending on image-related task. *Notes:* Model performance compared to VGG model (baseline).

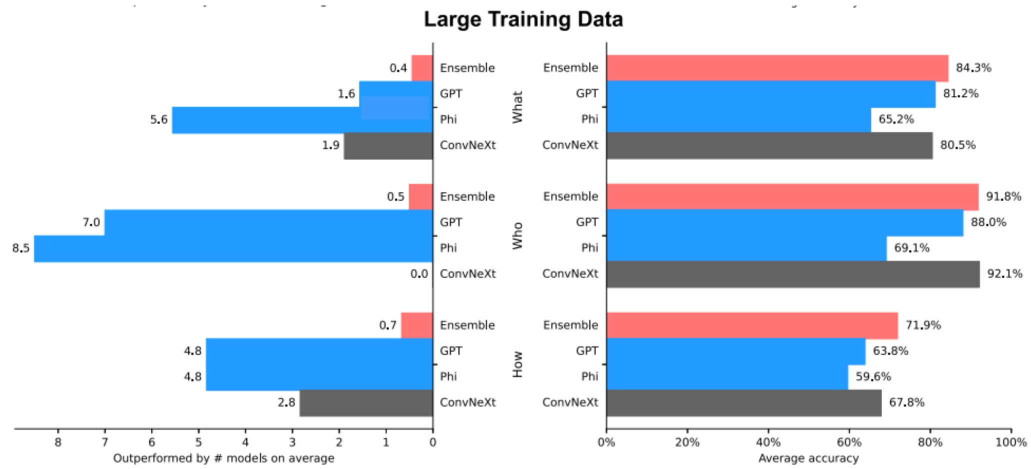


Fig. 6. Ensemble performance across all datasets. *Note:* GPT and Phi performance differs by training size because the large training data excludes the AIDA funnel car dataset.

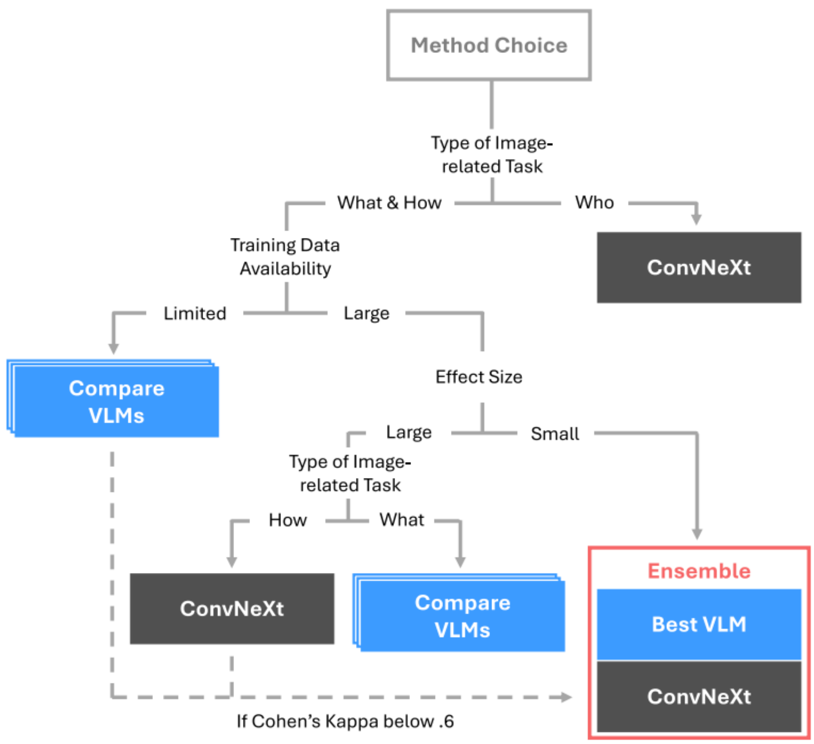


Fig. 7. Decision framework for method choice.

Table 1
Core categories for image classification and current marketing practices.

Category	Current Marketing Practice
Research Domains & DVs	• *Advertising & promotional effectiveness:* Ad effectiveness, CTR, purchase intention, sales (e.g., Feng et al., 2021; Hartmann et al., 2021) • *Brand management:* Brand perception, hiring outcome (e.g., Klostermann et al., 2018; Lee, 2021) • *Consumer insights & behavior:* Emotional response, engagement, viewer reaction (e.g., Chuah & Yu, 2021; Peng et al., 2020; Zhang et al., 2023) • *Media management:* Booking rate, intention to watch, popularity, sales (e.g., Liu et al., 2018; Schwenzow et al., 2021) • *Product design & management:* Aesthetic appeal, crowdfunding success, product rating, return rates, sales (e.g., Burnap et al., 2023; Dzyabura et al., 2023)
Image-related Tasks	• *What* is on the image (e.g., Hartmann et al., 2021; Overgoor et al., 2022; Zhang & Luo, 2022) • *Who* is on the image (e.g., Bharadwaj et al., 2022; Zhang et al., 2021) • *How* is the image perceived (e.g., Lee, 2021; Liu et al., 2020)
Dataset Characteristics	• *Number of training images:* Ranging from 203 to > 500,000 (Burnap et al., 2023; Zhang et al., 2021) • *Number of classes:* Ranging from 2 to > 99 (Chang et al., 2023; Zhang et al., 2023)
Methods	• *Full-NN-Training:* e.g., 4-layer CNN, 7-layer CNN, trained from scratch (Hu et al., 2019; Peng et al., 2020) • *Transfer learning:* e.g., VGG, ResNet, pretrained on ImageNet (Dzyabura et al., 2023; Hartmann et al., 2021)
Method Training	• *Hyperparameter selection:* e.g., batch size, epochs, learning rate (Hartmann et al., 2021; Liu et al., 2020) • *Evaluation:* e.g., train-validation-test split, 5-fold cross validation (Hartmann et al., 2021; Liu et al., 2020) • *Metrics:* e.g., accuracy, AUC-ROC (Burnap et al., 2023; Troncoso & Luo, 2023; Zhang et al., 2021)

Note: CTR = click-through rate, CNN = convolutional neural network, AUC-ROC = area under the curve of the receiver operating characteristic, analysis based on 49 publications in marketing-related journals (see Web Appendix A).

Table 2
Overview of all datasets in the empirical evaluation.

Dataset	Research domain	Image-related task	Number of images	Classes	Image dimensions	Source
AIDA Funnel Car	Advertising & Promotional Effectiveness	How (low, medium, or high AIDA)	559	3	341 × 283	Heitmann et al. (2025)
Brand Logo	Brand Management	What (e.g., electronics, foods, sports)	158,652	9	482 × 396	Wang et al. (2020)
Brand Selfie	Consumer Insights & Behavior	What (brand-, consumer selfie, or packshot)	6,928	3	481 × 491	Hartmann et al. (2021)
E-Commerce Category	Product Design & Management	What (e.g., parfum, smartphones, tools)	117,590	41	710 × 710	E-commerce Products (n.d.)
Emotion	Consumer Insights & Behavior	How (e.g., anger, joy, surpise)	8,350	6	722 × 555	Panda et al. (2018)
Face Sentiment	Consumer Insights & Behavior	How (savory, unsavory)	12,420	2	295 × 350	Good guys-Bad Guys (n.d.)
Fashion Products	Product Design & Management	What (e.g., bags, shoes, watches)	44,441	25	60 × 80	Fashion Product Images (n.d.)
Generated Aesthetics	Advertising & Promotional Effectiveness	How (low, medium, high)	10,320	3	512 × 512	Hartmann and Exner (2024)
Generated Ethnicity	Consumer Insights & Behavior	Who (e.g., asian, black, white)	10,001	4	256 × 256	Generated.photos (n.d.)
Generated Gender	Consumer Insights & Behavior	Who (female, male)	10,001	2	256 × 256	Generated.photos (n.d.)
Image Quality	Advertising & Promotional Effectiveness	How (e.g., low, medium, very high)	10,073	5	512 × 384	Hosu et al. (2020)
Image Sentiment	Consumer Insights & Behavior	How (e.g., highly positive, neutral, negative)	12,550	5	465 × 400	Image Sentiment Polarity (n.d)
Luxury Brands	Brand Management	What (e.g., Chanel, Gucci, Versace)	2,184	7	150 × 150	Brand Styles (n.d)
Outdoor Scenery	Media Management	What (e.g., buildings, forest, street)	17,034	6	150 × 150	Intel Scene Classification (n.d)
Public Places	Media Management	What (e.g., bathroom, conference room, kitchen)	9,456	21	700 × 700	Bylinskii et al. (2015)
Stock-photo Category	Media Management	What (e.g., arts & culture, business, nature)	8,000	16	200 × 221	Unsplash Images (n.d.)
Store Item Colour	Product Design & Management	What (e.g., black, red, yellow)	6,239	12	224 × 224	Store Items Color (n.d.)
Unbiased Emotion	Consumer Insights & Behavior	How (e.g., anger, fear, love)	3,045	6	837 × 659	Panda et al. (2018)

Note: Image dimensions are measured as the average of all images in the dataset and displayed as width × height.

Table 3
Step-by-step guide: evaluating vision language model performance.

Step	Description
1	**Define classification task:** • Specify type of image-related task (what/who/how classification) • Identify target classes (e.g., happy, sad, neutral)
2	**Prepare evaluation dataset:** • Create hold-out sample of 100-250 manually labeled images • Ensure sufficient observations of minority class for imbalanced datasets • Have multiple annotators label ambiguous cases to establish ground truth
3	**Choose models and design prompt:** • Identify relevant VLMs for evaluation (e.g., OpenAI GPT, Microsoft Phi, Meta Llama) • Create clear and specific instructions • Consider providing context or examples (e.g., few-shot prompting) • Test reasoning structures (e.g., chain-of-thought and role-based prompting)
4	**Run inference and calculate performance metrics:** • Run inference on entire hold-out dataset after estimating cost and runtime based on a sample • Compute accuracy (correct predictions / total predictions) and Cohen's Kappa • For imbalanced datasets, calculate precision, recall, F1-score, and ROC-AUC • Identify classes with poor performance through confusion matrix
5	**Assess marketing viability:** • Compare performance against predetermined threshold • Evaluate business impact of errors • Consider cost-benefit given zero training requirements
6	**Document edge cases:** • Identify systematic failure patterns • Note content restriction refusals • Record unexpected or nonsensical outputs
7	**Make implementation decision:** • Deploy if performance meets requirements • Use fine-tuned method if inadequate performance • Document rational for method selection in research

c. Authors' Conclusions

- These average observations suggest that VLMs can compete with fine-tuned models. When training data collection is costly or when researchers are faced with other data collection limitations, VLMs represent a low-cost, readily available alternative that researchers should take advantage of.
- VLMs can achieve high-quality predictions without task-specific fine-tuning.
- For who-related classification tasks or when large training datasets are available, ConvNeXt generally outperforms VLMs and provides a robust solution.
- When effect sizes are small, maximizing accuracy becomes critical; in such cases, researchers should compare multiple VLMs and implement an ensemble combining TVMs and VLMs.
- Conventional CNNs have become less relevant for marketing as modern language-based architectures offer superior performance.
- Findings demonstrate that VLMs have fundamentally changed image classification in marketing research. These developments make advanced image analysis more accessible to marketing researchers than ever and can improve understanding of how visual communication affects consumer behavior and market outcomes.

4. Discussion

a. Implications

- These observations suggest that marketing researchers should strive for at least 1000 manually annotated training images, whenever feasible.
- Training a full neural network is conceptually inferior as the richness of images requires more image labels than marketing can often collect.
- This is particularly relevant for few-shot applications of VLMs, where base model training data, prohibited tasks, and capabilities are opaque to applied marketing.
- Strengths and weaknesses of CNNs and TVMs may likewise vary by classification task, leading to possible differences in relative accuracy across applications.
- Methods that outperform on one dataset may underperform on another, making it essential to examine the stability of relative performance.
- It is essential to always evaluate VLM accuracy using manually annotated marketing images
- Lighthouse leaderboard results based on isolated applications are not representative of the broader set of marketing-related classification tasks, even when tasks might initially appear straightforward

b. Limitations

- VLMs can also hallucinate and create factually incorrect output that is unsupported by data. Their actual training material is seldom disclosed, so it is unclear which tasks are beyond their training and when they rely on out-of-scope reasoning
- An unknown set of human-related queries can be restricted in these models to prevent misuse
- Despite the breadth of datasets and methods included in this investigation, it cannot cover all conceivable image classification tasks
- Only two VLMs (GPT-5 and Phi-4) included in the main analysis because of cost and computational resources.
- Our empirical comparison focuses on image classification.

c. Future Directions

- Other VLMs exist and the performance and multi-modal reasoning capabilities will continue to advance (e.g., OpenAI, 2025c).
- Found a relatively small impact of alternative prompting strategies on the tasks we studied but the role of prompting is task dependent and should be further explored for individual applications.
- Combining VLMs with TVMs in an ensemble is also promising.
- Beyond classification tasks, many marketing applications are also interested in object detection (e.g., logos and other marketing assets) and in studying the impact of presence, size, and location, e.g., brand logos.
- Explore VLMs for multi-image and video scenarios that recent technological advances enable.

--- Page Break Here ---

Response Letter (~600 words)

Assessment of the study

Witte et al. (2026) provide a solid argument for how visual communication is central to marketing. They argue that while other visual methods like convolutional neural networks (CNNs) help in understanding visual impacts, they fail to capture complex contextual information, lack accuracy, and are not accessible. Witte et al, therefore, apply other visual methods such as transformer-based vision models (TVMs) and vision language models (VLMs) such as GPT-5 or Phi-4 to measure visual impacts. By conducting over 2,275 experiments across diverse datasets, the authors provide the first comprehensive benchmark that moves beyond simple CNN evaluations to include TVMs and VLMs.

Strengths

One notable strength of this paper is juxtaposing CNNs, TVMs, and VLMs methods and identifying which approach is better to measure visual impacts. This gives researchers the option to know which approach works best and under which circumstances to employ these methods. From the findings, "VLMs such as GPT-5 and Phi-4 provide accurate findings across a wide range of image-related tasks without requiring specific fine-tuning." Although they highlight the weaknesses of using VLMs, I find it a strength because they provide people with a complete picture of how these visual detection methods work. As the authors posit, a combination of TVMs and VLMs can overcome the challenges of using only VLM.

Weaknesses

Despite its rigor, the study has some notable limitations. As the authors acknowledge, the analysis relies on only two VLMs (GPT-5 and Phi-4), leaving a wide range of other models underexplored. Given the diversity in architectures, training data, and alignment strategies across VLMs, it is unclear whether the findings can be generalized. This raises the question of whether similar conclusions would emerge if additional models were included in the analysis. Expanding the study to incorporate a broader set of VLMs could help assess the robustness and external validity of the results, and potentially reveal variation in how different systems respond to the same tasks.

Second, the "black box" nature of commercial VLMs poses a significant threat to scientific replicability. The authors highlight that commercial providers implement safety guardrails that can unpredictably refuse to classify images for ethnicity or gender, which are often critical variables in marketing research. Because these guardrails are frequently updated without notice, a study conducted today may yield different outcomes if repeated just a few months later. This lack of stability not only complicates replication efforts but also raises concerns about methodological transparency and the reliability of findings derived from such systems.

Other notes

After reading this article, as the authors highlighted throughout the paper, there is no one-size-fits-all approach in analyzing images or even texts. Each model comes with its own strengths and limitations, which makes careful selection and evaluation essential. As the authors highlight, VLMs should be applied with caution, particularly given their potential for inconsistency or error in certain contexts. This underscores the importance of incorporating human validation to identify and correct discrepancies. Rather than relying solely on automated outputs, a hybrid approach that combines model efficiency with human judgment may lead to more accurate and reliable results.